

EE 446: TinyML

Lab 1: Edge Impulse Implementation of a Simple DNN for Classification

Summer 2026

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University of Washington



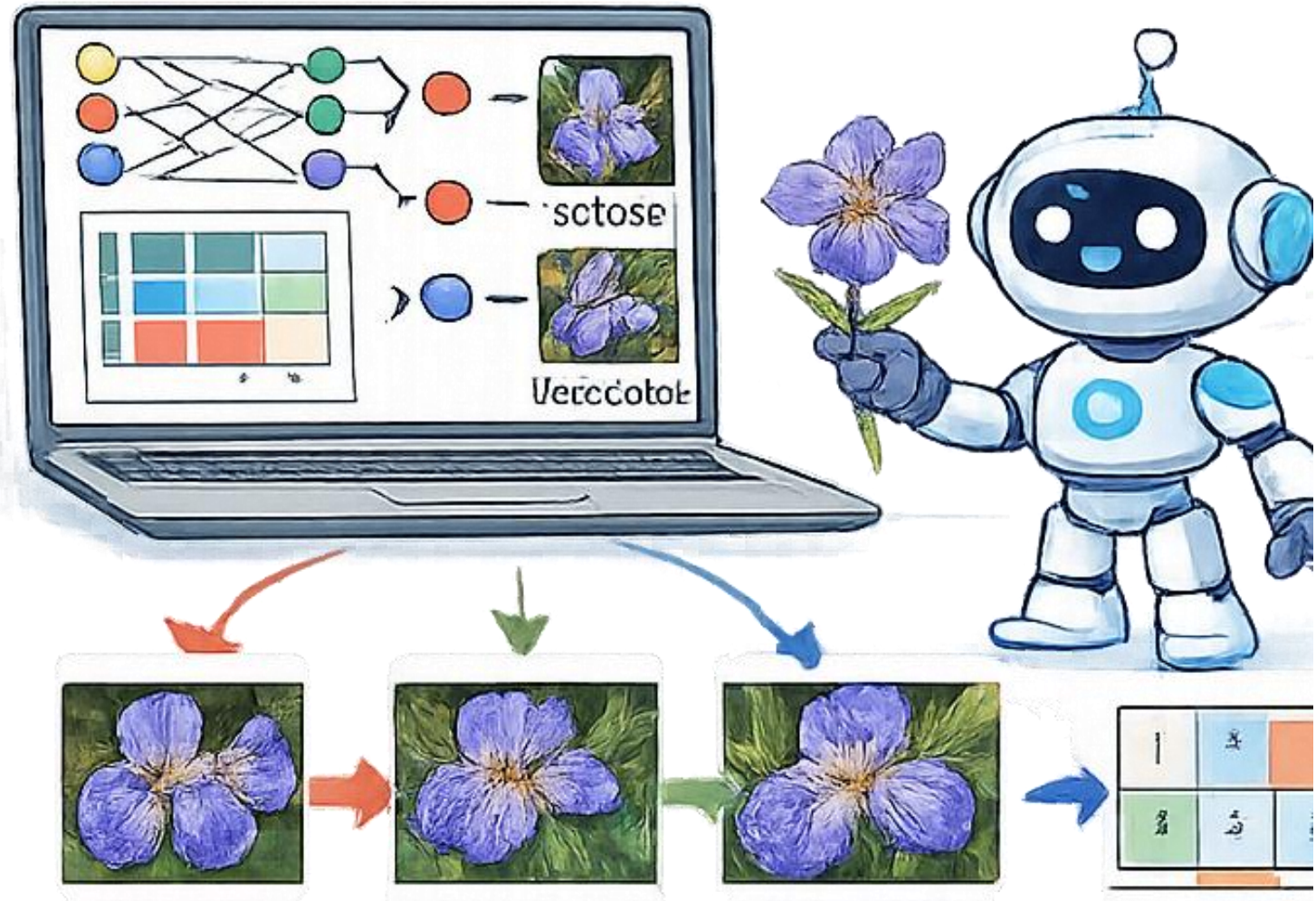
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UNIVERSITY of WASHINGTON



Lab #1

- Software Lab
- Create an Edge Impulse account and a new project
- Upload and inspect the normalized IRIS CSV dataset
- Build, train, and test a simple DNN classifier
- Interpret model accuracy and the confusion matrix



Lab 1 Overview

- **Goal:** Classify IRIS flower species using a simple neural network on Edge Impulse
- **Steps:**
 - Create an Edge Impulse account
 - Upload and process the IRIS dataset (in .csv file format)
 - Train and test a small DNN



EDGE IMPULSE

iris setosa



petal sepal

iris versicolor



petal sepal

iris virginica

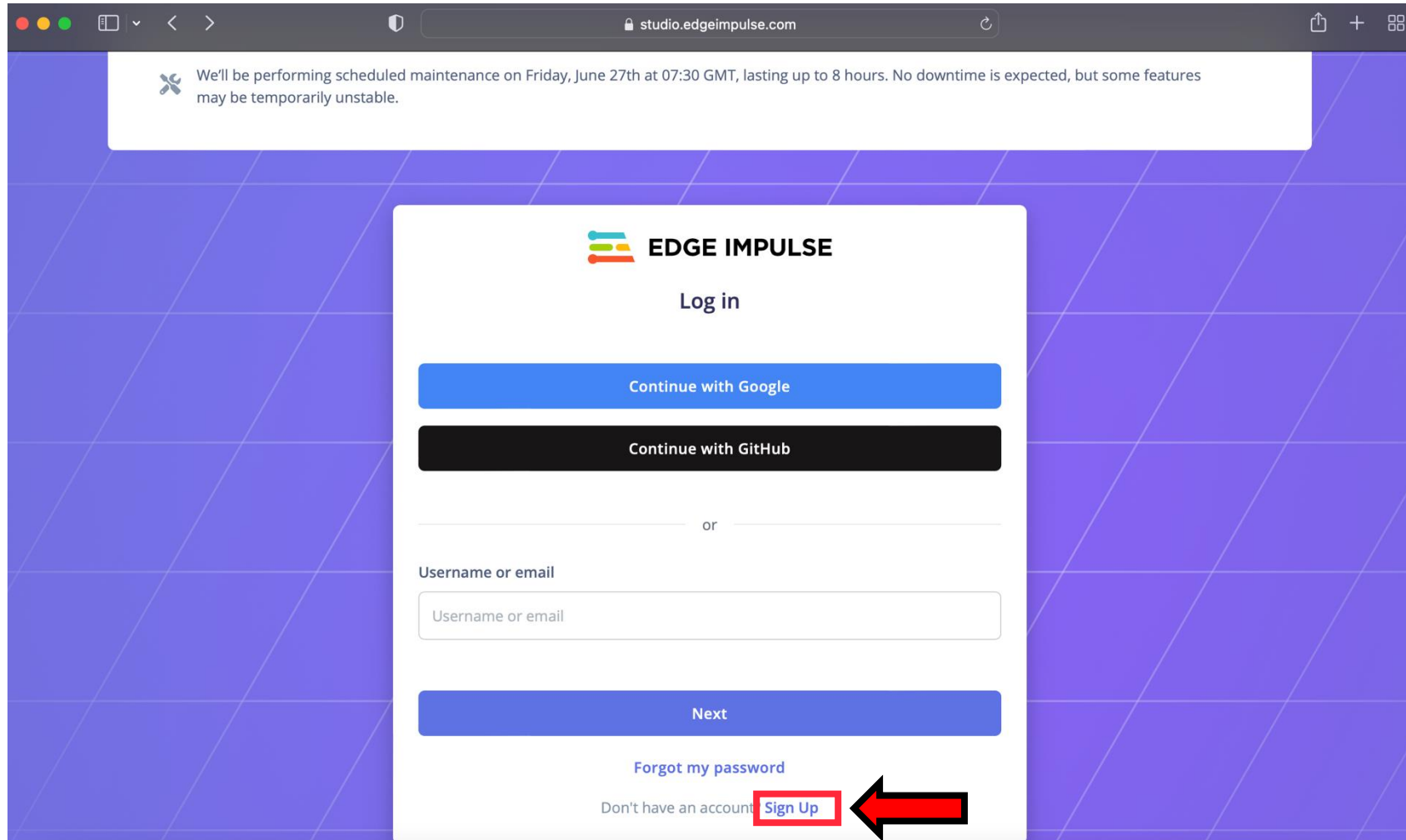


petal sepal

Source: <https://codesignal.com/learn/courses/intro-to-unsupervised-machine-learning/lessons/exploring-and-visualizing-the-iris-dataset>

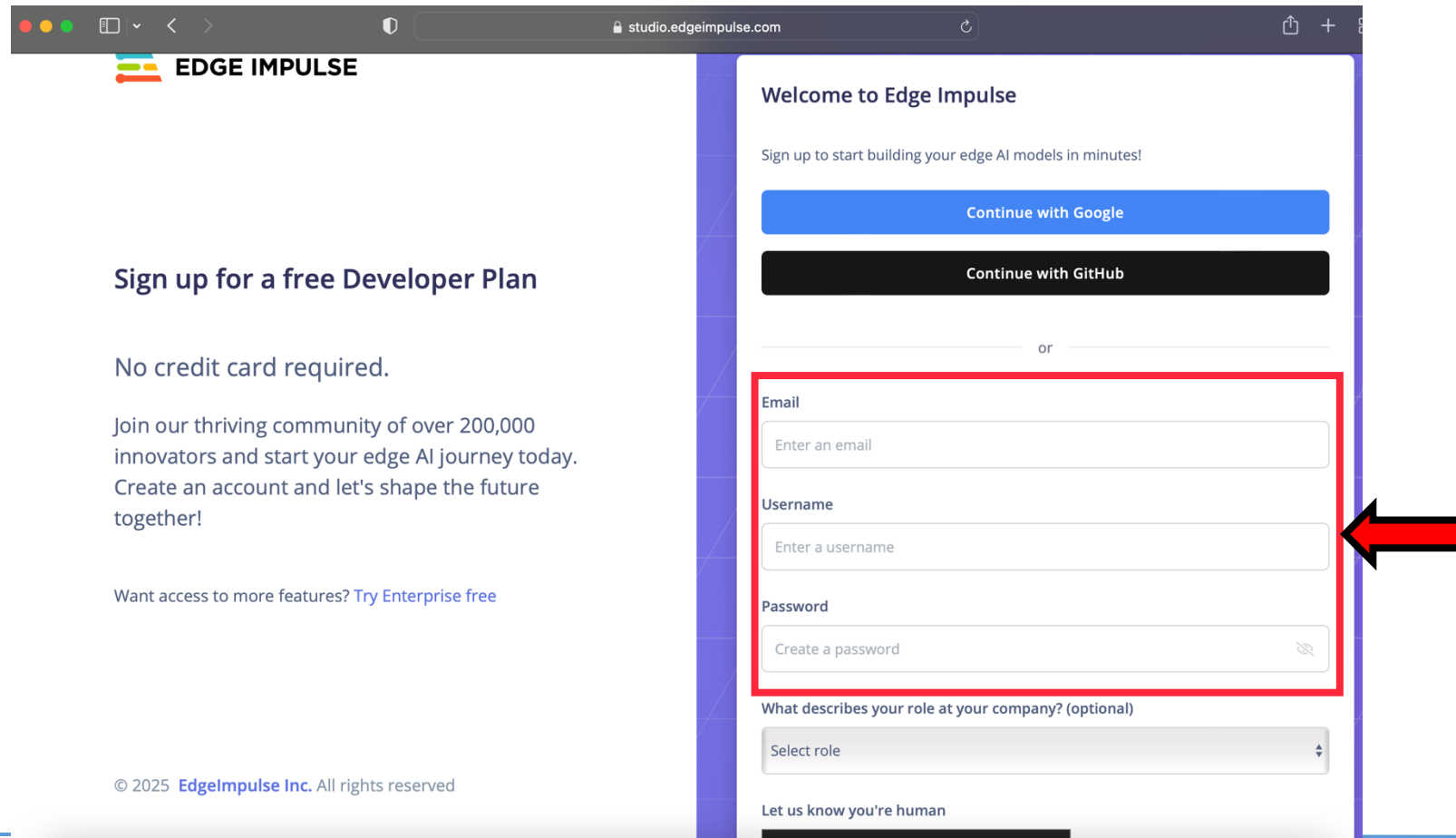
Step 0: Create an Edge Impulse Account

- Go to: <https://studio.edgeimpulse.com/>
- Click **Sign Up**



Step 0: Create an Edge Impulse Account

Use **UW student email (.edu)** (Preferred) and setup a username and password to create your Edge Impulse account.



The screenshot shows the Edge Impulse sign-up page in a web browser. The browser's address bar shows 'studio.edgeimpulse.com'. The page has a dark header with the 'EDGE IMPULSE' logo. The main content area is white with a blue sidebar on the left. The sidebar contains the text: 'Sign up for a free Developer Plan', 'No credit card required.', 'Join our thriving community of over 200,000 innovators and start your edge AI journey today. Create an account and let's shape the future together!', and 'Want access to more features? [Try Enterprise free](#)'. The main content area has a 'Welcome to Edge Impulse' heading, followed by 'Sign up to start building your edge AI models in minutes!'. There are two buttons: 'Continue with Google' (blue) and 'Continue with GitHub' (black). Below these is a red-bordered sign-up form. The form has three input fields: 'Email' (placeholder: 'Enter an email'), 'Username' (placeholder: 'Enter a username'), and 'Password' (placeholder: 'Create a password'). A red arrow points to the 'Username' field. Below the form is a dropdown menu labeled 'What describes your role at your company? (optional)' with the text 'Select role'. At the bottom of the form is a checkbox labeled 'Let us know you're human'.

studio.edgeimpulse.com

EDGE IMPULSE

Welcome to Edge Impulse

Sign up to start building your edge AI models in minutes!

Continue with Google

Continue with GitHub

or

Email

Enter an email

Username

Enter a username

Password

Create a password

What describes your role at your company? (optional)

Select role

Let us know you're human

Step 1: Create a new Project in Edge Impulse

- Create a new project: e.g., "IRIS Classification"

The screenshot shows the Edge Impulse Studio interface. On the left, a sidebar displays the user profile for Dinuka Sahabandu, a Developer, with a 'DS' icon and a 'DEVELOPER' badge. Below the profile, there's a section for 'Organizations' and a note about multi-factor authentication (MFA). The main area is titled 'Create a new project' and contains a form with three sections: 'Enter the name for your new project:' with a text input field containing 'IRIS Classification'; 'Choose your project type:' with radio buttons for 'Personal' (selected) and 'Enterprise'; and 'Choose your project setting:' with radio buttons for 'Public' (selected) and 'Private (1 of 3 remaining)'. A green 'Create new project' button is at the bottom of the form. A red box highlights the entire form area. In the background, the main interface shows a '+ Create new project' button in the top right corner, which is also highlighted with a red box and labeled 'Step 1'. A red arrow labeled 'Step 2' points from the 'Create new project' button in the background to the 'Create new project' button in the foreground form. Another red arrow labeled 'Step 3' points from the 'Create new project' button in the foreground form to the 'Create new project' button in the background.

Step 2: Download IRIS Dataset

- Download the CSV file from **Canvas Couse Page** → **Files**:
`iris_normalized_dataset.csv`

The screenshot displays the Canvas LMS interface for the course 'E E 400 A'. The left sidebar contains navigation links: Home, Modules, Syllabus, Announcements, Assignments, Discussions, Quizzes, Grades, Pages, Collaborations, Panopto Recordings, People, **Files** (highlighted with a red box and labeled 'Step 1'), Ally Course, Accessibility Report, Poll Everywhere, and Namecoach. The main content area shows the 'Files' section for 'E E 400 A Su 25: Advan'. A search bar and '0 items selected' are at the top. Below is a table of files:

Name	Date Created	Date Modified	Modified By	Size
iris_normalized_data	3:32am	3:32am		12 KB
template resources	Oct 5, 2017			--

A red box highlights the three-dot menu for 'iris_normalized_data' (labeled 'Step 2'). A red box highlights the 'Download' option in the dropdown menu (labeled 'Step 3').

Step 2: Details of IRIS Dataset

- Create a new project: e.g., "IRIS Classification"
- 150 samples, 3 classes:
 - Setosa
 - Versicolor
 - Virginica
- 4 features:
 - Sepal length, sepal width
 - Petal length, petal width
- All features normalized to [0, 1] for easier training

Data
Samples

4-d feature vector $x \in \mathbb{R}^4$ Label $y \in \{0, 1, 2\}$

	A	B	C	D	E
1	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)	label
2	0.22222222	0.625	0.06779661	0.04166667	setosa
3	0.16666667	0.41666667	0.06779661	0.04166667	setosa
4	0.11111111	0.5	0.050847458	0.04166667	setosa
5	0.08333333	0.45833333	0.084745763	0.04166667	setosa
6	0.19444444	0.66666667	0.06779661	0.04166667	setosa
7	0.30555556	0.79166667	0.118644068	0.125	setosa
8	0.08333333	0.58333333	0.06779661	0.08333333	setosa
9	0.75	0.5	0.627118644	0.54166667	versicolor
10	0.58333333	0.5	0.593220339	0.58333333	versicolor
11	0.72222222	0.45833333	0.661016949	0.58333333	versicolor
12	0.33333333	0.125	0.508474576	0.5	versicolor
13	0.61111111	0.33333333	0.610169492	0.58333333	versicolor
14	0.38888889	0.33333333	0.593220339	0.5	versicolor
15	0.55555556	0.54166667	0.627118644	0.625	versicolor
16	0.16666667	0.16666667	0.389830508	0.375	versicolor
17	0.63888889	0.375	0.610169492	0.5	versicolor
18	0.61111111	0.5	0.694915254	0.79166667	virginica
19	0.58333333	0.29166667	0.728813559	0.75	virginica
20	0.69444444	0.41666667	0.762711864	0.83333333	virginica
21	0.38888889	0.20833333	0.677966102	0.79166667	virginica
22	0.41666667	0.33333333	0.694915254	0.95833333	virginica
23	0.58333333	0.5	0.728813559	0.91666667	virginica
24	0.61111111	0.41666667	0.762711864	0.70833333	virginica
25	0.94444444	0.75	0.966101695	0.875	virginica
26	0.94444444	0.25	1	0.91666667	virginica
27	0.47222222	0.08333333	0.677966102	0.58333333	virginica
28	0.72222222	0.5	0.796610169	0.91666667	virginica

Step 3: Upload Dataset to Edge Impulse Project

EDGE IMPULSE

Dashboard

Devices

Data acquisition

Experiments

EON Tuner

Impulse design

Create impulse

Retrain model

Live classification

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Dinuka Sahabandu / IRIS Classification PERSONAL

Target: Cortex-M4F 80MHz DS

Dataset

Data explorer

Data sources

Synthetic data

AI labeling

CSV Wizard

CSV Wizard - Step 1: Upload a file

CSV Files come in many shapes and forms. Upload either:

An example CSV, TXT or Parquet file.

Or, an existing CSV Wizard config (you can obtain this from the CSV Wizard page on another project).

Choose File

iris_normalized_dataset.csv

Upload file

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Step 3: Upload Dataset to Edge Impulse Project



Dashboard

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CSV Wizard - Step 2: Process "iris_normalized_dataset.csv"

Here are the first 200 lines of your CSV file. You should see clear tabular data with named columns.

Delimiter: Comma (,)

Skip first lines: 0

No header row: ☐

ROW #	SEPAL LENGTH (CM)	SEPAL WIDTH (CM)	PETAL LENGTH (CM)	PETAL WIDTH (CM)	LABEL
# 1	0.222222222222221	0.625	0.06779661	0.041666667	setosa
# 2	0.1666666666666674	0.4166666666666674	0.06779661	0.041666667	setosa
# 3	0.1111111111111116	0.5	0.050847458	0.041666667	setosa
# 4	0.083333333	0.4583333333333326	0.084745763	0.041666667	setosa
# 5	0.1944444444444442	0.666666666666667	0.06779661	0.041666667	setosa
# 6	0.305555555555556	0.791666666666665	0.11864406779661013	0.125	setosa
# 7	0.083333333	0.5833333333333333	0.06779661	0.083333333	setosa
# 8	0.1944444444444442	0.5833333333333333	0.084745763	0.041666667	setosa
# 9	0.027777778	0.375	0.06779661	0.041666667	setosa

[Different format?](#)

Looks good, next



Step 3: Upload Dataset to Edge Impulse Project



- Dashboard
- Devices
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- EON Tuner
- Impulse design
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CSV Wizard - Step 3: About your data

Let's learn a bit about the structure of your data.

Is this time-series data?

- ☐ Yes, this is time-series data (either raw sensor data, or processed features)
- ☒ No

[Different format?](#)

Great, let's look at your values

iris_normalized_dataset.csv

ROW #	SEPAL LENGTH (CM)	SEPAL WIDTH (CM)	PETAL LENGTH (CM)
# 1	0.222222222222221	0.625	0.06779661
# 2	0.1666666666666674	0.4166666666666674	0.06779661
# 3	0.11111111111111116	0.5	0.050847458
# 4	0.083333333	0.4583333333333326	0.084745763
# 5	0.1944444444444442	0.666666666666667	0.06779661
# 6	0.305555555555556	0.7916666666666665	0.11864406779661013
# 7	0.083333333	0.5833333333333333	0.06779661
# 8	0.1944444444444442	0.5833333333333333	0.084745763
# 9	0.027777778	0.375	0.06779661
# 10	0.1666666666666674	0.4583333333333326	0.084745763
# 11	0.305555555555556	0.7083333333333333	0.084745763
# 12	0.1388888888888884	0.5833333333333333	0.1016949152542373
# 13	0.1388888888888884	0.4166666666666674	0.06779661
# 14	0	0.4166666666666674	0.016949153



Step 4: Upload Dataset to Edge Impulse Project



- Dashboard
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 - Retrain model
 - Live classification


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Dinuka Sahabandu / IRIS Classification PERSONAL

 Target: Cortex-M4F 80MHz

DS

Dataset | Data explorer | Data sources | Synthetic data | AI labeling | **CSV Wizard**

< CSV Wizard - Step 4: About your values

Almost there. Let's parse your values.

Do you have a column that contains the label (the value you want to predict)?

☒ Yes, it's

☐ No (no worries, you can provide this when you upload data)

Which columns contain your values?

☒ sepal length (cm)

☒ sepal width (cm)

☒ petal length (cm)

☒ petal width (cm)

[Different format?](#)

Finish wizard

Example samples (one per row)

LABEL (LABEL)	SEPAL LENGTH (CM)	SEPAL WIDTH (CM)	PETAL LENGTH (CM)
setosa	0.222222222222221	0.625	0.06779661
setosa	0.1666666666666674	0.4166666666666674	0.06779661
setosa	0.1111111111111116	0.5	0.050847458
setosa	0.083333333	0.4583333333333326	0.084745763
setosa	0.1944444444444442	0.666666666666667	0.06779661
setosa	0.3055555555555556	0.7916666666666665	0.1186440677966
setosa	0.083333333	0.583333333333333	0.06779661

Step 1

Step 2



Step 4: Upload Dataset to Edge Impulse Project

EDGE IMPULSE

Dataset | Data explorer | Data sources | Synthetic data | AI labeling | **CSV Wizard**

← CSV Wizard - Step 6: Finished

Any CSV files that you upload into your project - whether it's through the uploader, the CLI, the API or through data sources - will now be processed according to the rules you set up here.

Upload some data

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Impulse design

- Create impulse
- Retrain model
- Live classification

Step 4: Upload Dataset to Edge Impulse Project

You can upload CBOR, JSON, CSV, Parquet, WAV, JPG, PNG, AVI or MP4 files. You can also upload an annotation file named "info.labels" with your data to assign bounding boxes, labels, and/or metadata. View [Uploader docs](#) to learn more. Alternatively, you can use our [Python SDK](#) to programmatically ingest data in various formats, such as pandas or numpy; or [import data directly from another project](#).

CSV and Parquet files will be processed using the active CSV Wizard configuration, which was created using "iris_normalized_dataset.csv" (Today, 04:00:53). [Download config](#).

Upload mode

- ☒ Select individual files ?
- ☐ Select a folder ?

Select files

[Choose Files](#) iris_normalized_dataset.csv

Upload into category

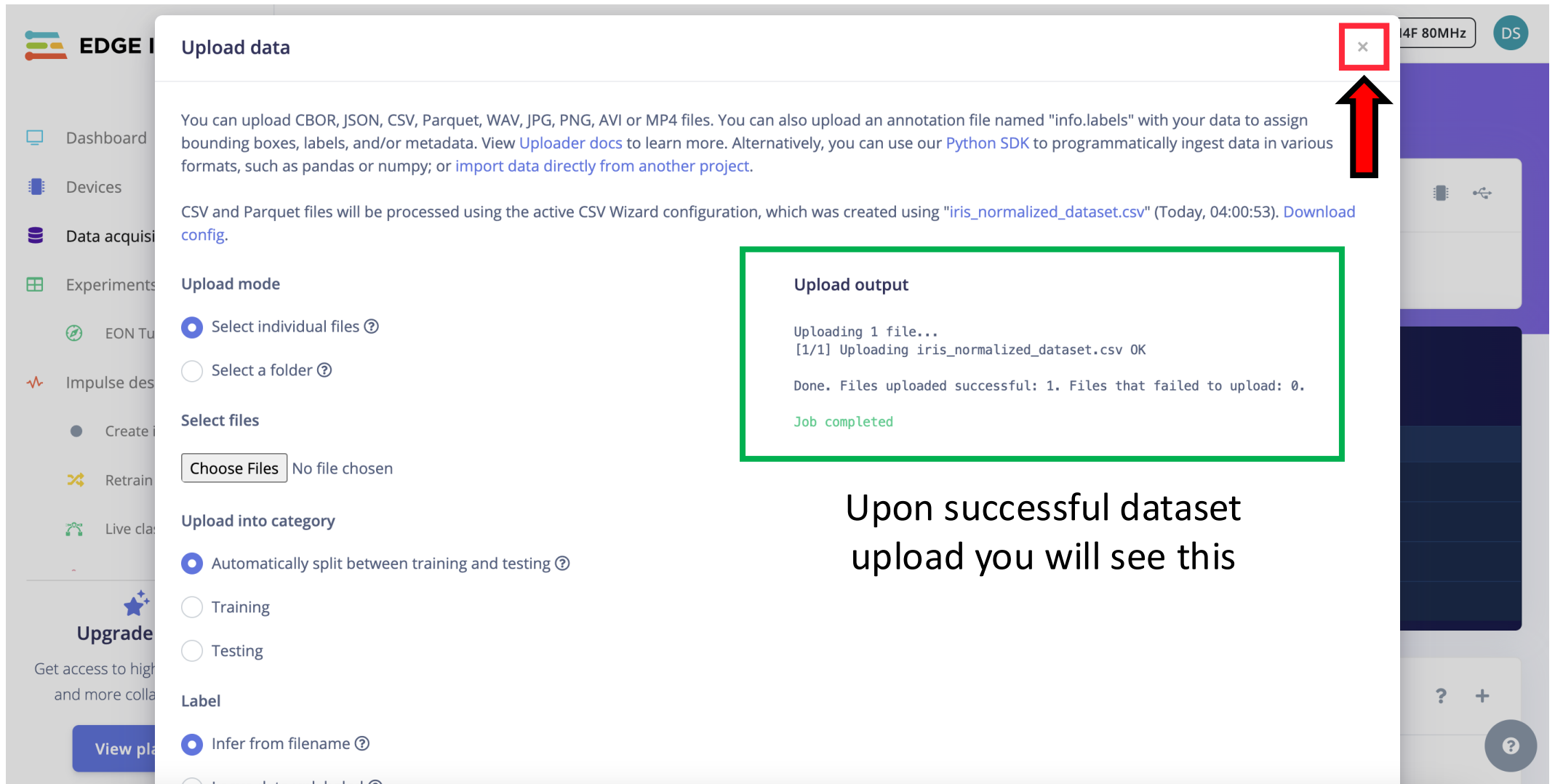
- ☒ Automatically split between training and testing ?
- ☐ Training
- ☐ Testing

Label

? CSV and Parquet files will be labeled using the 'label' column, as configured in the CSV wizard. [Update configuration](#)

Upload data

Step 4: Upload Dataset to Edge Impulse Project



The screenshot shows the 'Upload data' dialog box in the Edge Impulse web interface. The dialog is titled 'Upload data' and contains instructions on supported file formats (CBOR, JSON, CSV, Parquet, WAV, JPG, PNG, AVI, MP4) and the ability to upload annotation files. It also mentions that CSV and Parquet files are processed using the active CSV Wizard configuration. The 'Upload mode' section has two options: 'Select individual files' (selected) and 'Select a folder'. The 'Select files' section shows a 'Choose Files' button and 'No file chosen'. The 'Upload into category' section has three options: 'Automatically split between training and testing' (selected), 'Training', and 'Testing'. The 'Label' section has two options: 'Infer from filename' (selected) and 'Leave data unlabeled'. A green-bordered box highlights the 'Upload output' section, which displays the following text: 'Uploading 1 file...', '[1/1] Uploading iris_normalized_dataset.csv OK', 'Done. Files uploaded successful: 1. Files that failed to upload: 0.', and 'Job completed'. A red arrow points to a close button (X) in the top right corner of the dialog box.

Upon successful dataset upload you will see this

Step 5: Create Impulse

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AI labeling

CSV Wizard

DATA COLLECTED

2m 29s

TRAIN / TEST SPLIT

80% / 20%

Collect data

Connect a device to start building your dataset.

Dataset

Training (119) Test (30)

SAMPLE NAME	LABEL	ADDED	LENGTH
iris_normalized_d...	virginica	Today, 04:04:...	1s
iris_normalized_d...	virginica	Today, 04:04:...	1s
iris_normalized_d...	virginica	Today, 04:04:...	1s
iris_normalized_d...	virginica	Today, 04:04:...	1s
iris_normalized_d...	virginica	Today, 04:04:...	1s
iris_normalized_d...	virginica	Today, 04:04:...	1s
iris_normalized_d...	virginica	Today, 04:04:...	1s
iris_normalized_d...	virginica	Today, 04:04:...	1s
iris_normalized_d...	virginica	Today, 04:04:...	1s

RAW DATA

iris_normalized_dataset.s145

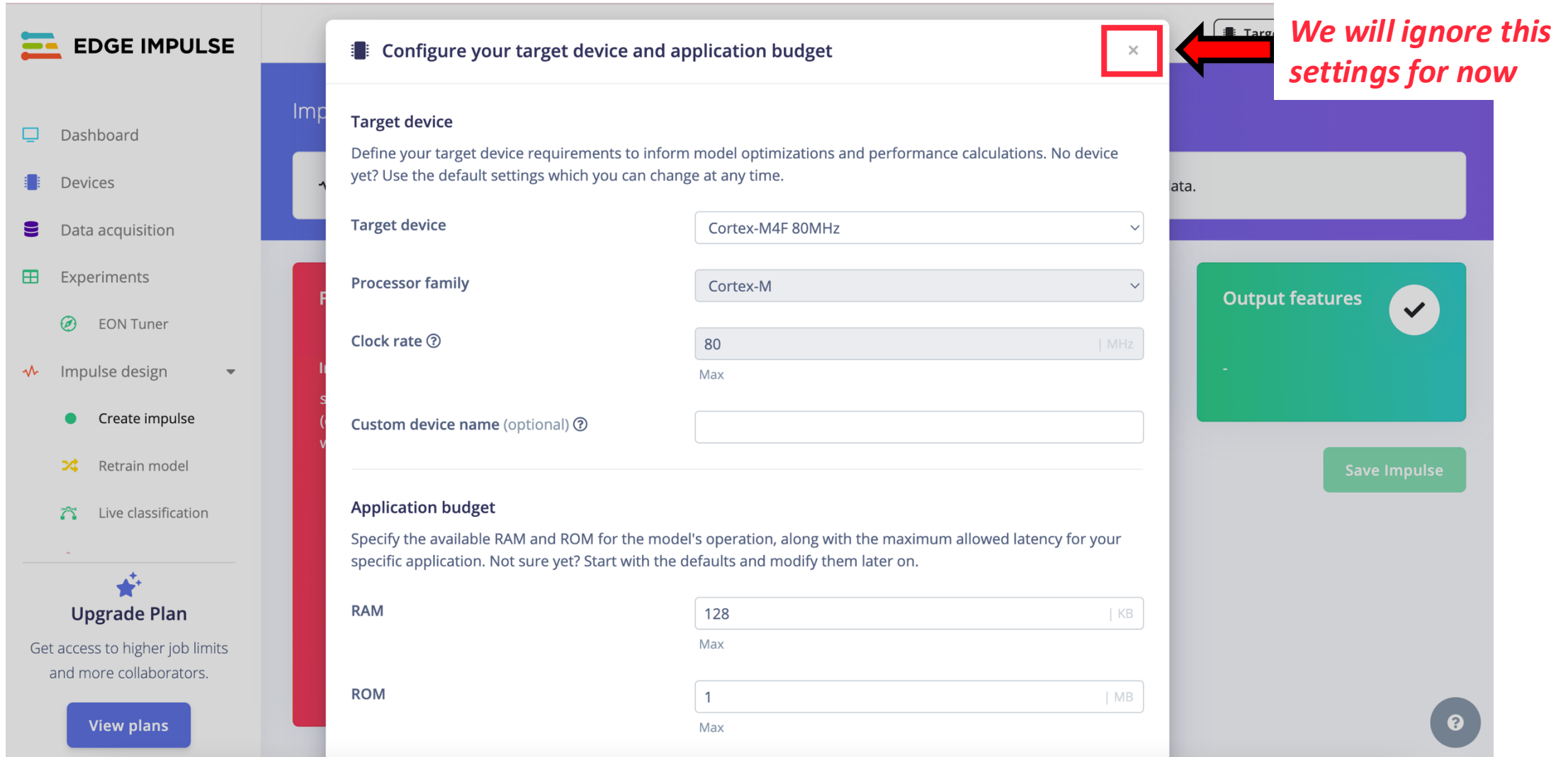
KEY	VALUE
sepal length (cm)	0.6666666666666667
sepal width (cm)	0.5416666666666665
petal length (cm)	0.7966101694915254
petal width (cm)	1

Metadata

No metadata.



Step 5: Create Impulse



EDGE IMPULSE

- Dashboard
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Configure your target device and application budget

Target device
Define your target device requirements to inform model optimizations and performance calculations. No device yet? Use the default settings which you can change at any time.

Target device: Cortex-M4F 80MHz

Processor family: Cortex-M

Clock rate: 80 MHz

Custom device name (optional):

Application budget
Specify the available RAM and ROM for the model's operation, along with the maximum allowed latency for your specific application. Not sure yet? Start with the defaults and modify them later on.

RAM: 128 KB

ROM: 1 MB

Output features: ☒

[Save Impulse](#)

We will ignore this settings for now

Step 5: Create Impulse

EDGE IMPULSE

Dinuka Sahabandu / IRIS Classification PERSONAL Target: Cortex-M4F 80MHz DS

Impulse #1

An impulse takes raw data, uses signal processing to extract features, and then uses a learning block to classify new data.

Features
Input features (4)
sepal length (cm), sepal width (cm), petal length (cm), petal width (cm)

Add a processing block

Add a learning block

Output features

Save Impulse

Step 5: Create Impulse

The screenshot shows the Edge Impulse web interface. On the left is a sidebar with navigation links: Dashboard, Devices, Data acquisition, Experiments, EON Tuner, Impulse design, Create impulse, Retrain model, and Live classification. The main area displays a project titled 'Dinuka Sahabandu / IRIS Classification' with a 'PERSONAL' tag. A modal dialog titled 'Add a processing block' is open in the center. It contains a table with the following data:

DESCRIPTION	AUTHOR	RECOMMENDED
Raw Data OFFICIALLY SUPPORTED Use data without pre-processing. Useful if you want to use deep learning to learn features.	Edge Impulse	★

A red box highlights the 'Raw Data' row, and a red arrow points to the 'Add' button in the 'RECOMMENDED' column. Below the table, there is a link 'Show all blocks anyway' and a button 'Add custom block'. The dialog also has a 'Cancel' button. In the background, the 'Output features' section is visible with a checkmark icon, and a 'Save Impulse' button is at the bottom right.

Step 5: Create Impulse

 **EDGE IMPULSE**

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Features

Input features (4)

sepal length (cm), sepal width (cm), petal length (cm), petal width (cm)

Raw Data

Name

Raw data

Input axes (4)

☒ sepal length (cm)

☒ sepal width (cm)

☒ petal length (cm)

☒ petal width (cm)

Add a learning block

Add a processing block

Save Impulse

An impulse takes raw data, uses signal processing to extract features, and then uses a learning block to classify new data.

Step 5: Create Impulse

EDGE IMPULSE

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Add a learning block

Did you know? You can [bring your own model](#) in PyTorch, Keras or scikit-learn.

DESCRIPTION	AUTHOR	RECOMMENDED
Classification OFFICIALLY SUPPORTED Learns patterns from data, and can apply these to new data. Great for categorizing movement or recognizing audio.	Edge Impulse	Add
Regression OFFICIALLY SUPPORTED Learns patterns from data, and can apply these to new data. Great for predicting numeric continuous values.	Edge Impulse	Add
Anomaly Detection (GMM) OFFICIALLY SUPPORTED Find outliers in new data. A Gaussian mixture model (GMM) models the shape of data using a probability distribution. New data that is unlikely according to this model can be considered anomalous.	Edge Impulse	Add
Anomaly Detection (K-means) OFFICIALLY SUPPORTED Find outliers in new data. Good for recognizing unknown states, and to complement classifiers. Works best with low dimensionality features like the output of the spectral features block.	Edge Impulse	Add

[Want access to all learning blocks? Upgrade now.](#)

Output features ✓
[Save Impulse](#)

Step 5: Create Impulse



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An impulse takes raw data, uses signal processing to extract features, and then uses a learning block to classify new data.

Features

Input features (4)
sepal length (cm), sepal width (cm), petal length (cm), petal width (cm)

Raw Data

Name
Raw data

Input axes (4)
☒ sepal length (cm)
☒ sepal width (cm)
☒ petal length (cm)
☒ petal width (cm)

Classification

Name
Classifier

Input features
☒ Raw data

Output features
3 (setosa, versicolor, virginica)

Output features

3 (setosa, versicolor, virginica)

Save Impulse

Add a processing block

Add a learning block

Step 6: Raw Data

 **EDGE IMPULSE**

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Impulse design

● Create impulse

● Raw data

● Classifier

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View plans

Parameters

Generate features

Raw data

Show: All labels iris_normalized_dataset.s14

KEY	VALUE
sepal length (cm)	0.6666666666666667
sepal width (cm)	0.5416666666666665
petal length (cm)	0.7966101694915254
petal width (cm)	1

Raw features

0.6667, 0.5417, 0.7966, 1.0000

Parameters

Scaling

Scale axes 1

Save parameters

DSP result

Processed features

0.6667, 0.5417, 0.7966, 1.0000

On-device performance

PROCESSING TIME

1 ms

PEAK RAM USAGE

16 Bytes

Step 6: Raw Data



- Dashboard
- Devices
- Data acquisition
- Experiments
- EON Tuner
- Impulse design
 - Create impulse
 - Raw data
 - Classifier

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Dinuka Sahabandu / IRIS Classification PERSONAL

Target: Cortex-M4F 80MHz DS

Parameters **Generate features**

Training set

Data in training set	1m 59s
Classes	3 (setosa, versicolor, virginica)
Calculate feature importance	☐

Generate features

Feature explorer

No features generated yet.

Feature generation output (0)

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Step 6: Raw Data

**EDGE IMPULSE**

 Dashboard

 Devices

 Data acquisition

 Experiments

 EON Tuner

 Impulse design

 Create impulse

 Raw data

 Classifier

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Training set

Data in training set1m 59s

Classes3 (setosa, versicolor, virginica)

Calculate feature importance☐

[Generate features](#)

Feature generation output (0)

```
[ 1/119] Creating windows from files...
[ 73/119] Creating windows from files...
[119/119] Creating windows from files...
Creating windows from files OK

Created 119 windows: setosa: 41, versicolor: 39, virginica: 39

Creating features...
[ 1/119] Creating features...
[119/119] Creating features...
Created features
Attached to job 34741976...

Job completed (success)
```

Upon feature generation you will see this

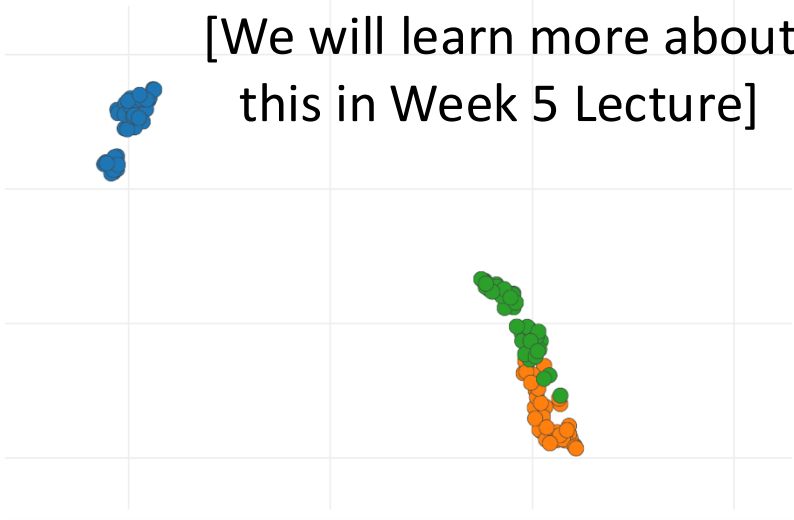
Feature explorer

 setosa

 versicolor

 virginica

This is visualization of 2d projection of 4d features [We will learn more about this in Week 5 Lecture]



On-device performance 

 **PROCESSING TIME**
0 ms.

 **PEAK RAM USAGE**
16 Bytes



Step 7: Classifier

EDGE IMPULSE

- Dashboard
- Devices
- Data acquisition
- Experiments
- EON Tuner
- Impulse design
 - Create impulse
 - Raw data
 - Classifier**

Upgrade Plan
Get access to higher job limits and more collaborators.
[View plans](#)

Neural Network settings

Training settings

- Number of training cycles ? 30
- Use learned optimizer ? ☐
- Learning rate ? 0.1
- Training processor ? CPU

Advanced training settings

Neural network architecture

- Input layer (4 features)
- Dense layer (20 neurons)
- Dense layer (10 neurons)
- Add an extra layer
- Output layer (3 classes)

Training output (0)

Step 1 →

Step 2 →

Step 3 (Delete these two layers) →

Here we are going to implement the simplest DNN model. Which is input (4 neurons) directly connecting to output layer (3 neurons)

Step 7: Classifier



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[View plans](#)

Neural Network settings

Training settings

Number of training cycles [?]

Use learned optimizer [?] ☐

Learning rate [?]

Training processor [?]

Advanced training settings

Neural network architecture

Input layer (4 features)

Add an extra layer

Output layer (3 classes)

[Save & train](#)

Training output

 (0)

Step 7: Classifier

The screenshot shows the Edge Impulse web interface. On the left is a sidebar with navigation links: Dashboard, Devices, Data acquisition, Experiments, EON Tuner, and Impulse design (which is expanded). Under 'Impulse design', there are three options: Create impulse, Raw data, and Classifier. At the bottom of the sidebar is an 'Upgrade Plan' section. The main area displays a list of layer types for a neural network model. The layers listed are: Dense, 1D Convolution / pooling, 2D Convolution / pooling, Reshape, Flatten, and Dropout. The 'Flatten' layer is highlighted with a red rectangular box, and a red arrow points to its 'Add' button. The descriptions for each layer are as follows:

LAYER TYPE
Dense Fully connected layer, the simplest form of a neural network layer. Use this for processed data, such as the output of a spectral analysis DSP block.
1D Convolution / pooling Learn features that take spatial information into account along a single dimension. Use this for raw data, or for DSP blocks that output spatial data, such as the MFCC block.
2D Convolution / pooling Learn features that take spatial information into account along two dimensions. Use this for raw data, or for DSP blocks that output spatial data, such as the MFCC block.
Reshape Turn one-dimensional data from a DSP block into multi-dimensional data. Use this as an input to a convolutional layer. Use this for deep learning on raw data, or to process MFCC output.
Flatten Flatten multi-dimensional data into a single dimension. You need to flatten data from a convolutional layer before returning.
Dropout Reduce the risk of a model overfitting your dataset by randomly cutting a fraction of network connections during training. Can be helpful if your model's training performance is better than its validation performance.

Step 7: Classifier

 **EDGE IMPULSE**

 Dashboard

 Devices

 Data acquisition

 Experiments

 EON Tuner

 Impulse design

Create impulse

Raw data

Classifier

 Upgrade Plan

Get access to higher job limits and more collaborators.

View plans

Training settings

Number of training cycles ?

30

Use learned optimizer ?

☐

Learning rate ?

0.1

Training processor ?

CPU

Advanced training settings

Neural network architecture

Input layer (4 features)

Flatten layer

Add an extra layer

Output layer (3 classes)

Save & train

Step 7: Classifier

Devices

Data acquisition

Experiments

EON Tuner

Impulse design

Create impulse

Raw data

Classifier

Retrain model

Live classification

Model testing

Deployment

Upgrade Plan

Get access to higher job limits and more collaborators.

View plans

Training settings

Number of training cycles 30

Use learned optimizer

Learning rate 0.1

Training processor CPU

Advanced training settings

Neural network architecture

Input layer (4 features)

Flatten layer

Add an extra layer

Output layer (3 classes)

Save & train

Calculating performance metrics...
Calculating inferencing time...
INFO: Created TensorFlow Lite XNNPACK delegate for CPU.
Calculating inferencing time OK
Calculating float32 accuracy...
INFO: Created TensorFlow Lite XNNPACK delegate for CPU.
Calculating int8 accuracy...

Model training complete
Attached to job 34743182...
Attached to job 34743182...
Attached to job 34743182...

Job completed (success)

Upon classifier training you will see this

Model

Model version: Unoptimized (float32)

Last training performance (validation set)

ACCURACY 95.8%

LOSS 0.22

Confusion matrix (validation set)

	SETOSA	VERSICOLOR	VIRGINICA
SETOSA	100%	0%	0%
VERSICOLOR	0%	100%	0%
VIRGINICA	0%	12.5%	87.5%
F1 SCORE	1.00	0.92	0.93



Step 8: Model Testing/Inference

Step 1 → Model testing

Step 2 → Classify all

This lists all test data. You can manage this data through [Data acquisition](#).

Test data

Set the 'expected outcome' for each sample to the desired outcome to automatically score the impulse.

SAMPLE NAME	EXPECTED OUTC...	ACCURACY	RESULT
iris_normali...	virginica		...
iris_normali...	virginica		...
iris_normali...	virginica		...
iris_normali...	virginica		...
iris_normali...	virginica		...
iris_normali...	virginica		...

Model testing output (0)

- **Test Set:** Data samples that have NOT been used in classifier training.
- Evaluation of test set allows us to evaluate model performance on unseen data (generalization capability)

<https://studio.edgeimpulse.com/studio/727530/impulse/1/classification>

Step 8: Model Testing/Inference

Devices

Data acquisition

Experiments

EON Tuner

Impulse design

Create impulse

Raw data

Classifier

Retrain model

Live classification

Model testing

Deployment

iris_normali...	virginica	100%	1 virginica	...
iris_normali...	virginica	100%	1 virginica	...
iris_normali...	virginica	0%	1 uncertain	...
iris_normali...	virginica	100%	1 virginica	...
iris_normali...	virginica	0%	1 uncertain	...

Job completed (success)

Results

Model version: ? Unoptimized (float32)

%

ACCURACY

70.00%

Metrics for Classifier

METRIC	VALUE
Area under ROC Curve ?	0.97
Weighted average Precision ?	0.90
Weighted average Recall ?	0.90
Weighted average F1 score ?	0.90

Confusion matrix

	SETOSA	VERSICOLOR	VIRGINICA	UNCERTAIN
SETOSA	100%	0%	0%	0%
VERSICOLOR	0%	45.5%	0%	54.5%
VIRGINICA	0%	10%	70%	20%
F1 SCORE	1.00	0.59	0.82	

Feature explorer ?

Upon classifier inference you will see this

Inference Performance



Lab Task 1

Complete the baseline model from implementation following Steps 1–8.

- Reproduce Slides 3–32 to build the baseline IRIS classifier in Edge Impulse.
- Include a screenshot of the **training performance** showing the **accuracy** and **confusion matrix**.
- Include a screenshot of the **inference / test performance** showing the **accuracy** and **confusion matrix**.
- Use the simple baseline architecture from Step 7.

Lab Task 2

Design and evaluate an improved classifier.

- Define a **classifier architecture that is different from Task 1.**
- Your **new model must achieve inference accuracy > 90%. If Task 1 already achieves > 90%, define a new architecture that achieves even higher inference accuracy than your Task 1 model.**
- Include a screenshot of your **neural network architecture.**
- Include screenshots of the **new training performance and the new inference / test performance**, each with **accuracy** and **confusion matrix**.

Lab Submission Guidelines

Make sure to organize your lab report in the following way:

- Prepare your lab report in Word or LaTeX.
- Put your **name and student ID** at the top of the report.
- Include **the public shareable link to your Edge Impulse project that hosts Lab 1.**
- Include **screenshots for Task 1 and Task 2.**
- **Export the report as a PDF.**
- **Submit the PDF via Canvas.**

NOTE: No points are awarded for a missing Edge Impulse public link.